UNIT-3

Define Virtualization,Need of Virtualization,Virtualization Technologies, Uses of Virtualization,Planning for Virtualization,Virtualization Pitfalls.

Define Virtualization

Virtualization is fundamentally an approach to pooling and sharing technology resources to simplify management and increase asset use. This approach ensures that IT resources can more readily meet business demand.

- For servers or networks, virtualization is used to take a single physical asset and make it operate as if it were multiple assets.
- This improves asset utilization and efficiency, and it decreases costs by reducing the need for physical assets.
- With storage or networks, virtualization acts as an abstracted view of underlying physical devices.
- This allows multiple physical assets to be combined and presented to servers and applications as if they were a single, larger asset.
- This dramatically simplifies server and application architecture and reduces costs.
- For desktops, virtualization is used to centralize the management of data and applications to reduce administration costs and data risk.

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Need of Virtualization

Virtualization has transitioned from an efficiency tool to being the core way IT gets done. Four primary trends and several operational needs drive its popularity:

1. **Underutilized hardware:** Prior to virtualization, many data centers had servers and storage running at 10 percent or less of total capacity. In other words, 90 percent of a device's potential was unused, representing a massive waste of resources. Virtualization separates the logical representation from the actual underlying physical devices. By applying virtualization, organizations can raise their hardware utilization rates from 10 or 15 percent to 70 or 80 percent. It breaks the one-to-one relationship of physical assets and creates pools of IT components that can be dynamically allocated.
2. **Data centers run out of space:** Businesses are far more computing-intensive today, with processes moving from paper-based to software systems. The rise of the Internet means massive increases in e-mail, websites, video, and mobile applications. Consequently, companies are running out of space in their data centers. Virtualization offers the ability to host multiple guest systems on a single physical server, allowing organizations to reclaim data center territory and avoid the tens of millions of dollars required to construct new spaces.
3. **Energy costs go through the roof:** Power costs are no longer assumed to be cheap and endlessly available. The cost of running hardware at low utilization rates is high;

virtualization makes more efficient use of compute resources to significantly reduce the overall cost of energy.

4. **IT operations costs mount:** Computers require constant care by system administrators to monitor status, replace hardware, install OS and software, patch applications, and support end-users. IT operations costs have risen with computing resource growth. Virtualization reduces operations costs by centralizing systems and providing a more flexible infrastructure.
5. **Virtualization Increases Business Agility:** Business conditions change rapidly, and organizations must respond instantly. Old processes associated with physical hardware provisioning made it difficult to react quickly. Virtualization makes it much quicker to spin up computing resources; tasks that took days or weeks can now be accomplished in minutes. It allows dynamic provisioning to support applications when capacity is needed most.
6. **Virtualization Increases IT Operational Flexibility:** Responding to hardware failures can take hours or days to replace hardware and bring a machine back online. Virtualization software keeps track of virtual machines and, if one goes down, immediately restarts another instance on the same or a different machine.
7. **Virtualization Reduces IT Operations Costs:** Traditionally, maintenance and upgrades were scheduled for off-hours, and sometimes skipped due to scheduling difficulties, leading to hardware failures. Virtualization enables a running virtual machine to be migrated to another server very quickly, freeing up the original server for work. By using fewer physical machines, hardware maintenance costs may be cut by 60 percent or more.
8. **Quality of Service:** Virtualization, backed by consistent management practices and software, ensures IT services avoid outages and uncoordinated activities.
9. **High Availability:** Also called HA, this acts as disaster or downtime avoidance. Server virtualization avoids planned and unplanned downtime by moving live, running virtual servers from affected hosts to other hosts.
10. **Disaster Recovery (DR):** When disaster strikes, keeping a second set of physical equipment ready but unused is wasteful and expensive. Virtual machines can be easily transferred within seconds or minutes to a backup data center on a smaller number of physical servers.
11. **End-User Manageability and Security:** Everyday desktop and laptop devices pose challenges for applying OS patches, application updates, and virus definitions. Client virtualization ensures client software images are maintained on centralized servers and pushed out to client devices. This also ensures critical corporate data is not stored on devices that can be lost or stolen.
12. **Virtualization Is Green:** Virtualization is ideally suited for companies wanting to reduce energy use and carbon footprints. Virtualization enables IT organizations to reduce the total number of machines and disk drives by up to 90 percent.

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Virtualization Technologies

Virtualization draws on well-established technology to provide several different flavors that abstract physical resources.

1. **Server Virtualization** There are three main types of server virtualization: operating system virtualization (containers), hardware emulation, and paravirtualization.

- **Hardware emulation:** The virtualization software, called a hypervisor, creates a virtual machine by emulating an entire hardware environment.
    - The operating system loaded into the VM is a standard, unmodified product.
    - As the OS makes calls for system resources, the hardware emulation software catches the system call and redirects it to manipulate data structures.
    - The hypervisor itself makes calls to the actual physical hardware.
    - This is often called bare metal virtualization because no software sits between the hypervisor and the metal of the server.
- **Paravirtualization:** This approach does not attempt to emulate a hardware environment in software.
    - Instead, a paravirtualization hypervisor multiplexes access to the underlying hardware resources of the server.
    - The hypervisor resides directly on the hardware.
    - A privileged guest runs as a guest virtual machine but has privileges that allow it to directly access certain resources on the underlying hardware.
- **Virtualization players:** VMware uses hardware emulation for its flagship platform, vSphere. Citrix offers XenServer which uses paravirtualization. Microsoft offers Hyper-V, whose architecture is similar to Xen but refers to virtual machines as partitions. Hewlett-Packard provides a broad portfolio of virtualization products across server, storage, network, and client hardware.

2. **Storage Virtualization** Storage virtualization is the process of abstracting logical storage from physical storage. The physical storage resources (such as disk drives) are aggregated into storage pools. Virtualization can be implemented within storage arrays or at the network level, pooling arrays from different vendors into a single monolithic storage device.

- **Network-attached storage (NAS):** A storage device on the network that offers storage to servers. It uses file-based protocols such as NFS or SMB/CIFS where computers request a file rather than a disk block. It is IP based and simple to deploy, commonly used for rich media, documents, and e-mail.
- **Storage area networks (SAN):** A storage device accessible to servers so the devices appear locally attached to the operating system. A SAN provides block-level operations rather than file abstractions. Most SANs use Fibre Channel connectivity or iSCSI. They are used for transactionally accessed data like databases and high usage file servers.

3. **I/O Virtualization** Getting data off the machine requires going through network and storage endpoints on the server. Manually managing these physical endpoints means an IT organization can only operate as efficiently as manual management allows. I/O virtualization makes these devices intelligent, allowing I/O context to be switched among physical devices.

4. **Network Virtualization** Virtualization technology is applied to the network itself so that network changes can be accomplished without moving cables between physical resources. Network virtualization allows the network to be reconfigured on the fly. Virtualization-capable network devices are managed remotely and configured logically.
5. **Client Virtualization**

- **Application virtualization:** This refers to the separation of program execution from program display. A program executes on a server in the data center, but the graphical output is sent to a remote client device. A variant streams the application to the client device each time it is fired up, ensuring the IT organization can keep it updated with versions and patches.
- **Desktop virtualization (VDI):** A user's entire PC executes on a central server, with the graphical display output to a client device. Instead of storing thousands of duplicate disk images, modern VDI uses a single image that is cloned as required, cutting down on storage. This often uses thin clients, which are inexpensive devices with little computing power and no local disk storage.
- **Desktop streaming:** In this check-in, check-out variant, ongoing storage is centralized, but the client system is transferred down to the client device for use. When the session ends, the image is written back to the repository and nothing remains on the hardware.

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Uses of Virtualization

Virtualization is applied across numerous use cases, ranging from simple to transformational.

1. **Studying Server Consolidation:** This is the most common first application of virtualization. Consolidation takes separate server instances and migrates them into virtual machines running on fewer servers. Companies often move from 150 physical servers to 150 virtual machines on only 15 servers, yielding a 60 to 80 percent utilization improvement and significant reductions in hardware, power, cooling, and software licensing costs.
    a. *Case Study:* First American Corporation virtualized 50 percent of its servers using VMware on HP BladeSystem servers. They run 4,500 open systems on 2,250 physical machines. They achieved a 98 percent reduction in space required and a 90 percent drop in power requirements, saving an estimated $714,000 annually.
2. **Development and Test Environments:** Developers must run and test software on a variety of systems and versions. Virtualization allows an engineer to replicate a distributed environment containing several systems on a single piece of hardware. Furthermore, when early software crashes, it only damages the specific application and its associated virtual machine, rather than destroying an entire physical machine's underlying operating system.
3. **Private Cloud Computing:** Cloud computing provides on-demand technology services over the Internet. A private cloud operates these services solely for a single organization. Virtualization is a key component because it allows for rapid provisioning and de-provisioning of these on-demand services.

4. **Quality of Service:** Virtualization removes hardware dependence, enabling organizations to quickly respond to hardware or network failures, or pre-emptively avoid failure by moving workloads.
    a. **Simple failover:** The hypervisor constantly monitors VM status and starts a new instance if a running VM is no longer present. Outage duration is mere seconds.
    b. **High availability (HA):** A crashed VM is started on a different hardware server to avoid hardware-precluding failures. Coordinating virtualization software monitors multiple hypervisors and handles the complex migration of the VM's network address and storage resources to the new physical server.
    c. **Clustering:** This ensures no data is lost during a failure. The coordinating software runs two identical VMs on separate machines. The primary server continually sends state changes to the secondary server. If the primary fails, users are transparently switched to the backup server without noticing.
    d. **Data mirroring:** Data in one place is reflected to another, enabling real-time consistency between two data stores.
    e. **Data replication:** Keeps complete copies of data available at a centralized location for system rebuild purposes.
5. **IT Operational Flexibility:** IT organizations must be agile to grow or shrink computing resources rapidly.
    a. **Load balancing:** Redundancy is achieved by running more than one copy of a virtual machine on separate servers. Both VMs carry half the load, ensuring resources aren't going unused.
    b. **Server pooling:** Virtualization software manages a pool of virtualized servers. The software automatically figures out which physical server is best suited to run a VM and migrates it as necessary to rebalance loads.
6. **Helping with Disaster Recovery:** Disaster recovery addresses catastrophic situations where an entire data center is lost. Keeping spare computing capacity in a remote data center is extremely expensive. With virtualization, a smaller set of machines can be kept available in a remote backup data center; in a disaster, VM images are transferred and started in just a few minutes.

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Planning for Virtualization

Embarking on a virtualization initiative requires significant structural planning and expert help, moving systematically through defined steps.

1. **Remember That Virtualization Is a Journey, Not a Product:** A successful virtualization journey requires preparation, multiple steps, and ongoing oversight to ensure a happy outcome. After organizations start receiving benefits, they tend to look for additional applications of the technology.
2. **Evaluate Your Use Cases:** A use case defines how you will currently, or might in the future, apply virtualization. Evaluating these helps identify capabilities you might need in the future, as the range of suitable products narrows as applications become more complex. You can utilize services like HP's Virtualization Workshop or Proof of Concept services to establish these baselines.

3. **Review Your Operations Organization Structure:** Virtualization is not purely technical; it impacts the human side of IT. Traditional IT operations have isolated groups for servers, networks, and storage. Virtualization integrates these functions, requiring these autonomous groups to collaborate and cooperate.
4. **Define Your Architecture:** Take your use cases and define your virtualization infrastructure design. Review this architecture with all interested parties to ensure everyone's needs are captured and to build commitment and momentum for the project.
5. **Select Virtualization Software:** Select products from major providers like VMware, Citrix, Microsoft, or HP that meet the defined architecture. Organizations should rethink infrastructure to manage virtual and physical resources with the same tools, using converged infrastructure software.
   a. *Case Study Application:* Terremark Worldwide Inc. used HP Network Automation software to manage a multi-tenant environment, achieving single tool multi-vendor support and cutting service setup times from 24-to-36 hours to mere minutes.
6. **Select Virtualization Hardware:** Ensure new hardware is virtualization-ready. Repurposing old hardware often fails because it cannot scale. Modern blades (like HP ProLiant) and shared storage systems (SAN or NAS) are strictly required to take advantage of advanced virtualization capabilities like live migration.
7. **Perform a Pilot Implementation:** Test your selected solution in a controlled test environment. Waiting until the production phase to find out if project assumptions work is the wrong approach.
8. **Implement Your Production Environment:** Order software and hardware, install data center equipment, configure virtualization tools, and rigorously test to confirm readiness to migrate to the new architecture. If the infrastructure isn't right, production systems will fail.
9. **Put Governance and Policy in Place:** Virtualization greatly reduces the effort required to implement new systems, which can lead to "virtual server sprawl". Avoid this by putting strict governance structures, formal rules, and policies in place.
10. **Manage Your Virtualized Infrastructure:** After implementation, you must administer the new environment. Integrate the virtual systems into overall management processes so physical and virtual systems are managed with the same tools and people. Organizations can also seek external IT outsourcing services to manage these environments.

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Virtualization Pitfalls

Navigating a virtualization project requires avoiding these known operational and strategic traps:

1. **Don't Wait for Perfection:** The virtualization field is in a great deal of flux and constantly releasing new products. Do not defer taking action just because you want to wait for the technology to become completely static.

2. **Don't Skimp on Training:** IT organizations often invest huge sums in hardware and software but skimp on ensuring their employees know how to use the new systems.
3. **Don't Expose Yourself to Legal Issues: Compliance:** When single physical systems are presented as multiple virtual systems, historical software licensing agreements can change. With vastly more systems generated through virtualization, tracking licenses is difficult but crucial. IT governance must be put in place to maintain compliance.
4. **Don't Imagine That Virtualization Is Static:** Virtualization technology is constantly changing; a state-of-the-art solution implemented 18 months ago must be continuously re-examined.
5. **Don't Skip the "Boring Stuff":** Installing software is fun, but conducting use case interviews and design reviews is mandatory. Without completing these "boring" tasks, you will likely not get the go-ahead to move forward with the project.
6. **Don't Overlook a Business Case:** Ignoring the business case is a sure way to get a project shot down. You must evaluate the financial impact of moving to virtualization and present this data to gain executive support and sail through the approval process.
7. **Don't Overlook the Organization:** Virtualization affects many groups; you must convince technical groups, business sponsors, and senior management that it will make their work easier. Track workflows explicitly so nothing falls between the cracks.
8. **Don't Overlook the Hardware:** Hardware dramatically impacts the virtualization density and performance levels available for virtual machines. Ensure that target hardware—whether blades, servers, or storage arrays—is completely capable of supporting the implementation.
9. **Don't Overlook Service Management:** Most virtualization platforms come with their own management tools, but they are not robust enough to manage beyond their intended platforms. Adopt solutions that manage physical and virtual environments in common, within an overall service framework.
10. **Be Prepared for Even More Virtualization in Your Future:** Virtualization reduces system creation time from days to mere minutes, which can cause organizations to suffer from a surfeit of systems sprawling everywhere. Keep an eye on the growth trends of your virtual machines. Virtualization will continually be applied to new areas of technology, so expect ever more of it in the future.